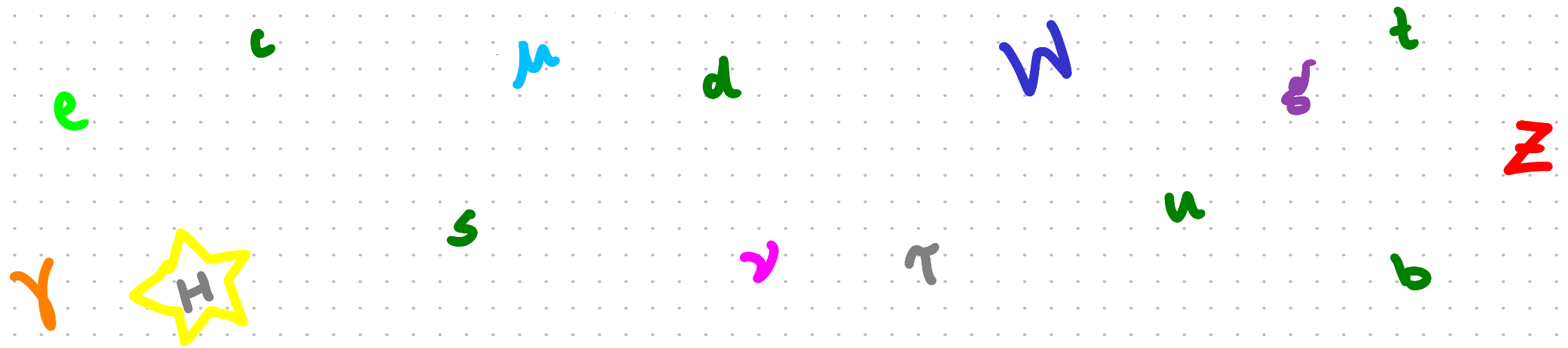
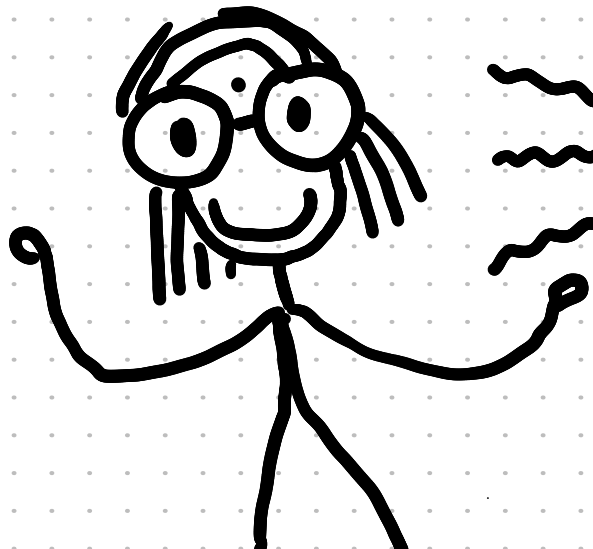




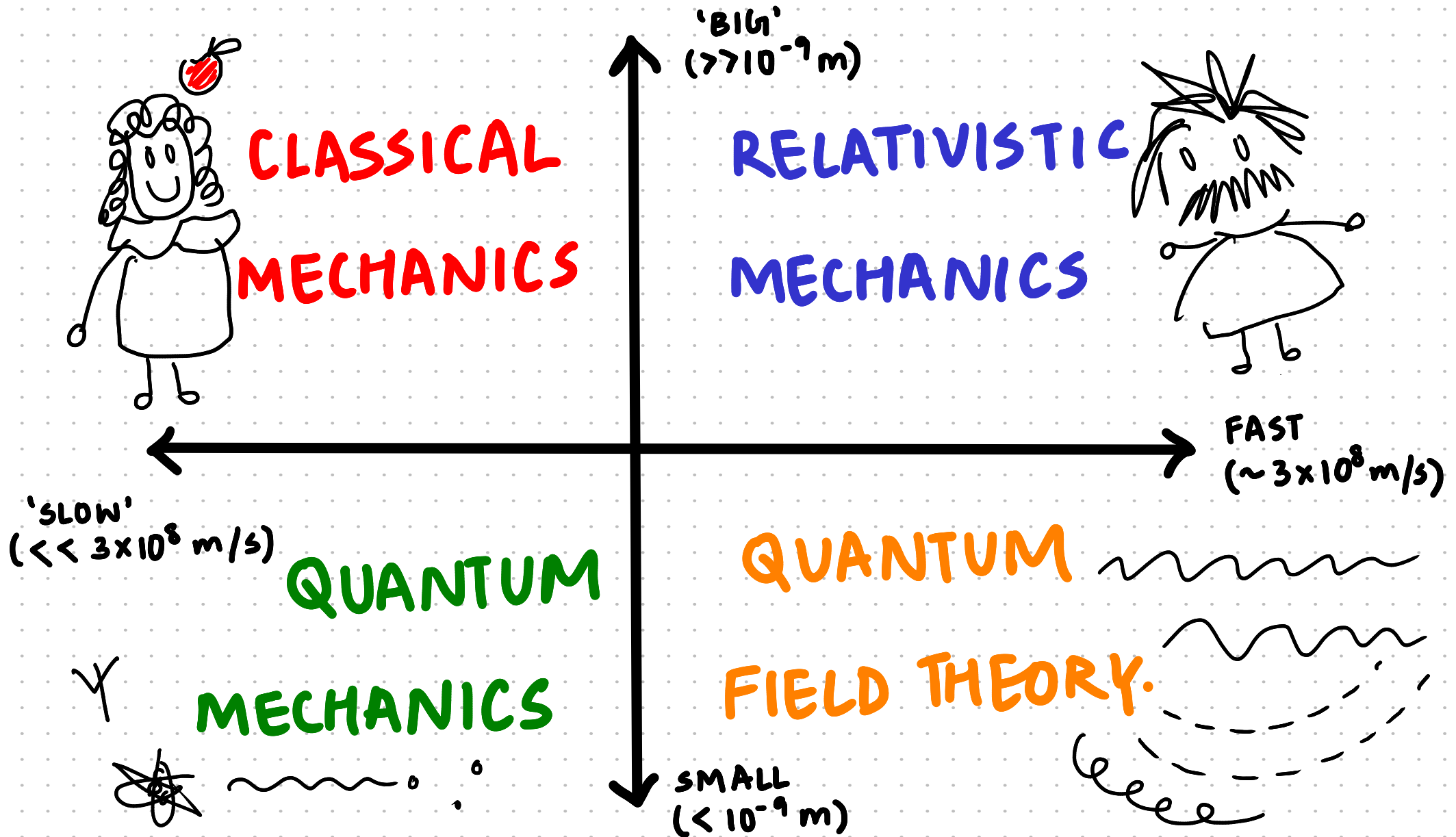
introduction to particle physics :)



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What do particle physicists study?

What is particle physics about?



ENERGY + MASS IN PARTICLE PHYSICS

DEALING
W/ V. SMALL
LENGTHS

$$\hbar = c = 1$$

TRAVELLING @
V. HIGH SPEEDS.

~~$$E = mc^2$$~~

$$E^2 = p^2 c^2 + m^2 c^4$$

→ E, m, p HAVE SAME UNIT

→ E, m, p MEASURED
IN eV.

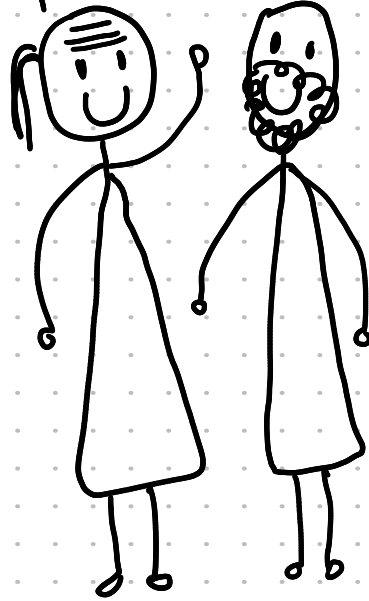
What is our universe made up of?

What is the most fundamental particle?

What is the most fundamental particle?

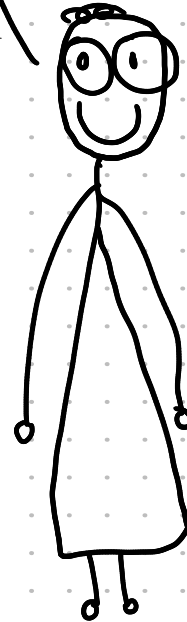
'INDIVISIBLE' PARTICLES!

5 NATURAL ELEMENTS!



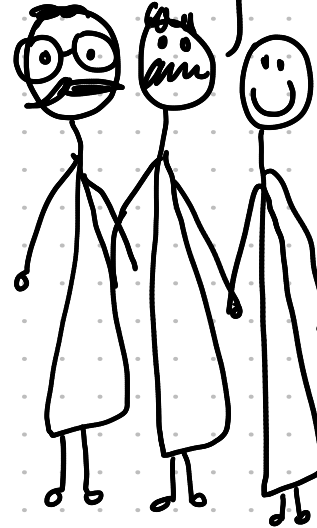
ANCIENT INDIA/GREECE

ATOMS!



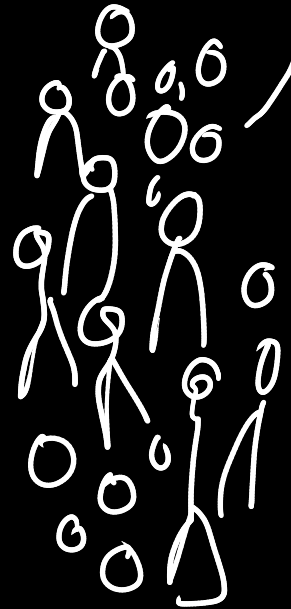
1800s - DALTON

PROTONS, NEUTRONS & ELECTRONS!



1900s - THOMSON, RUTHERFORD, CHADWICK.

WELL ACTUALLY.



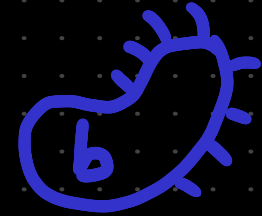
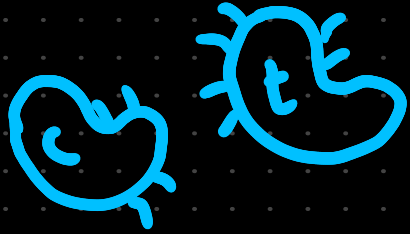
1900s - 2012 MODERN PHYSICISTS.

so far

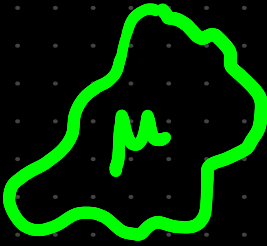
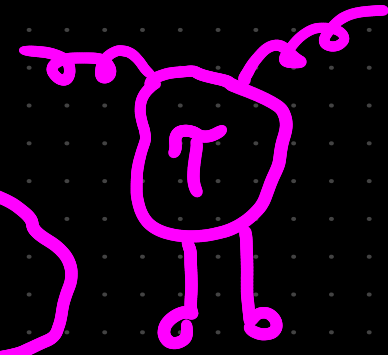
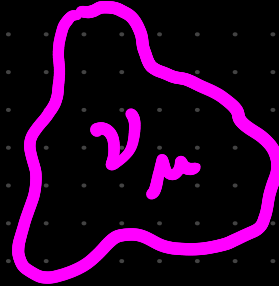
What is the most fundamental particle?

→ MOST FUNDAMENTAL
IS A MOVING
TARGET.

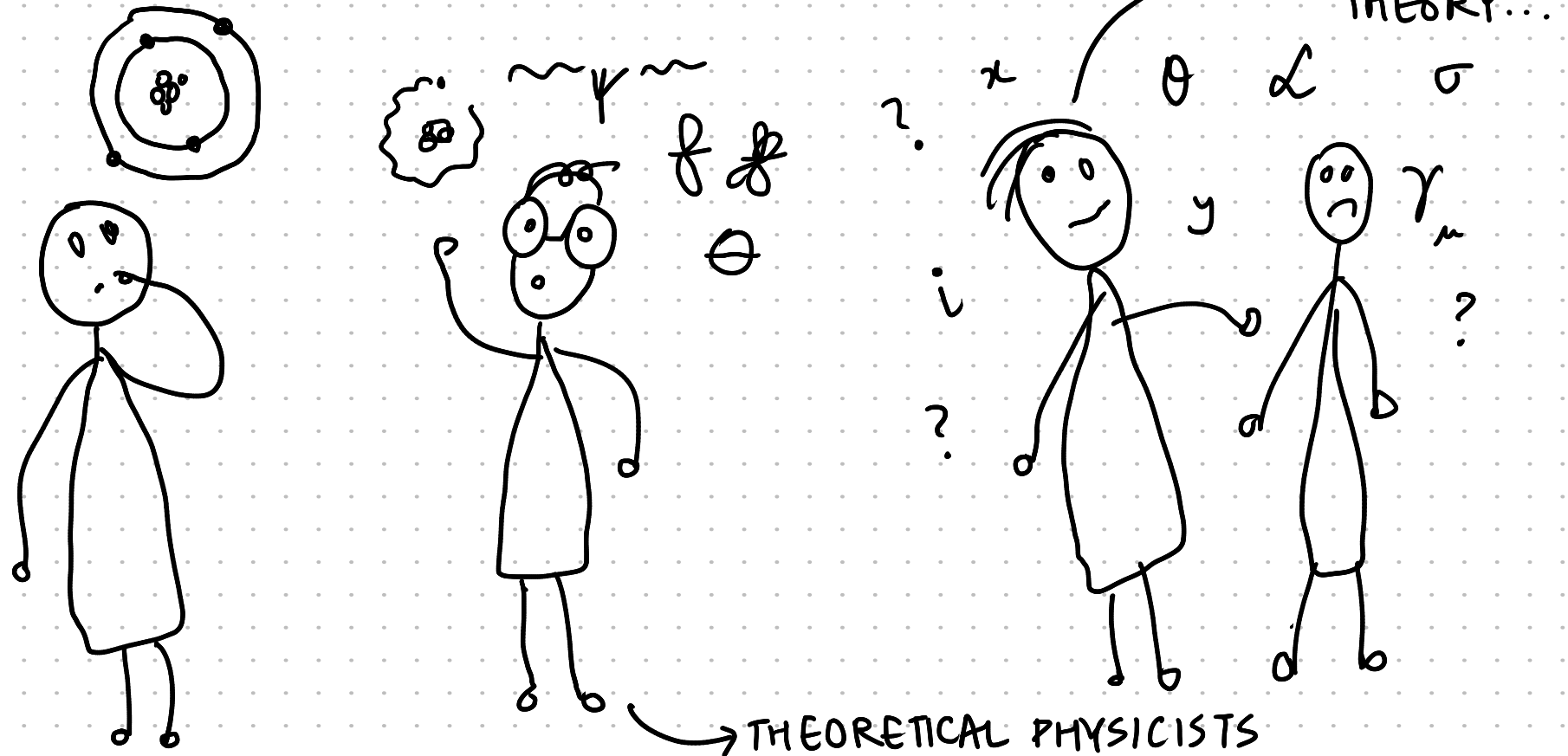
1930s - 90s

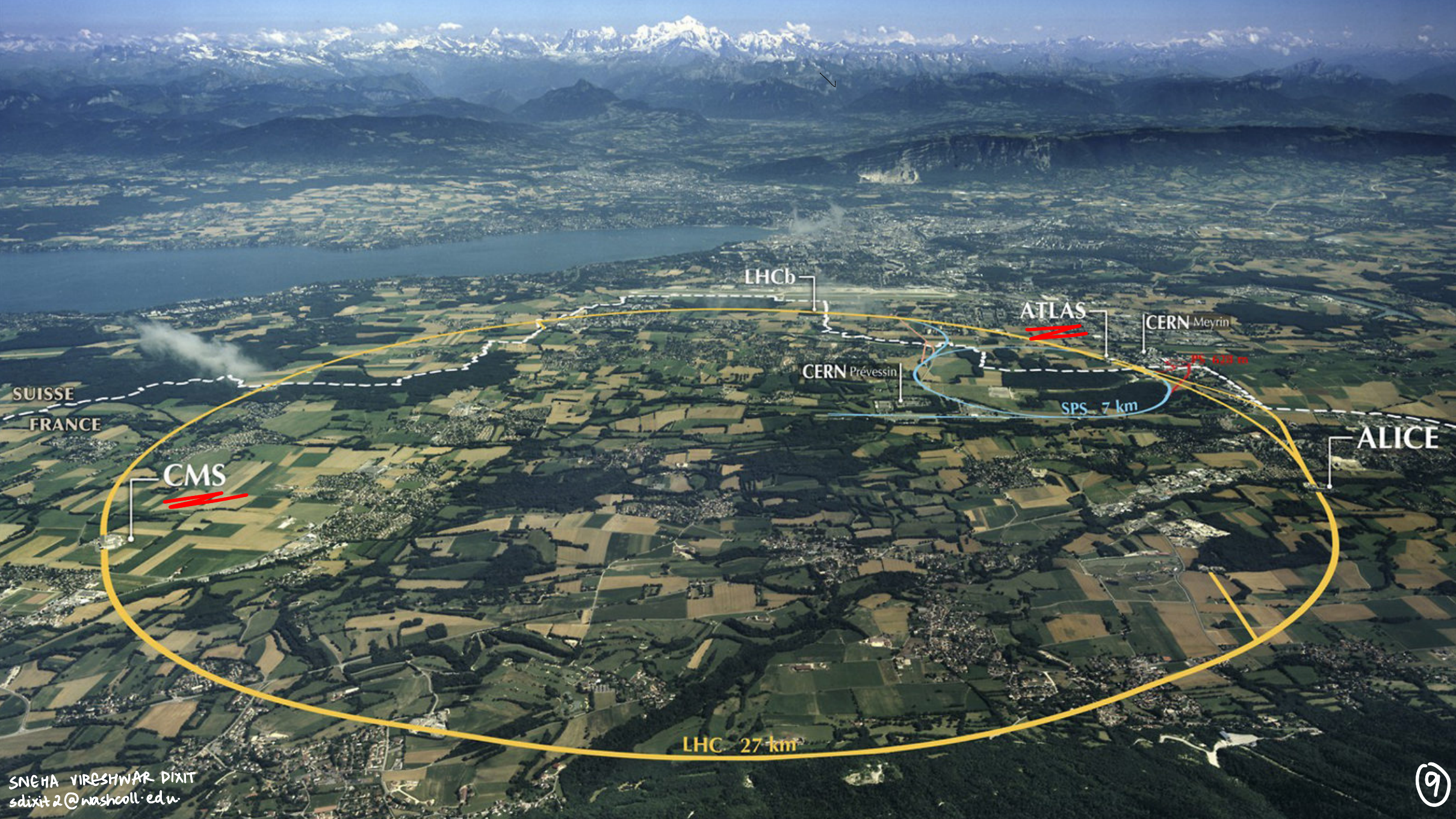


A PARTICLE ZOO
IS DISCOVERED!



How do we make sense of all these new particles?





LHCb

ATLAS

CERN Meyrin

PS 6.3 km

CERN Prévessin

SPS 7 km

ALICE

CMS

LHC 27 km

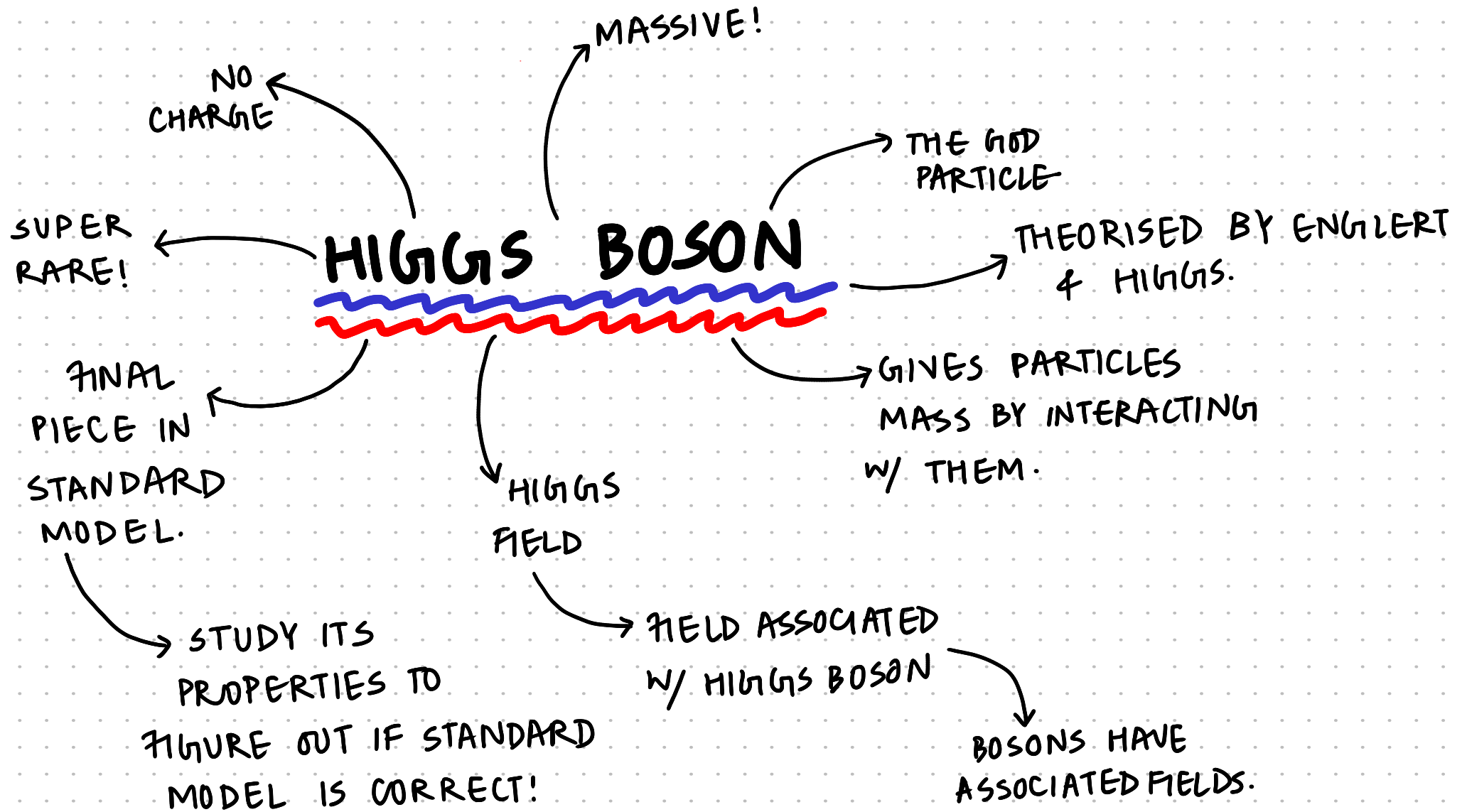
SUISSE
FRANCE

2012
CERN LHC.



→ 2013
NOBEL
(ENGLERT
& HIGGS)

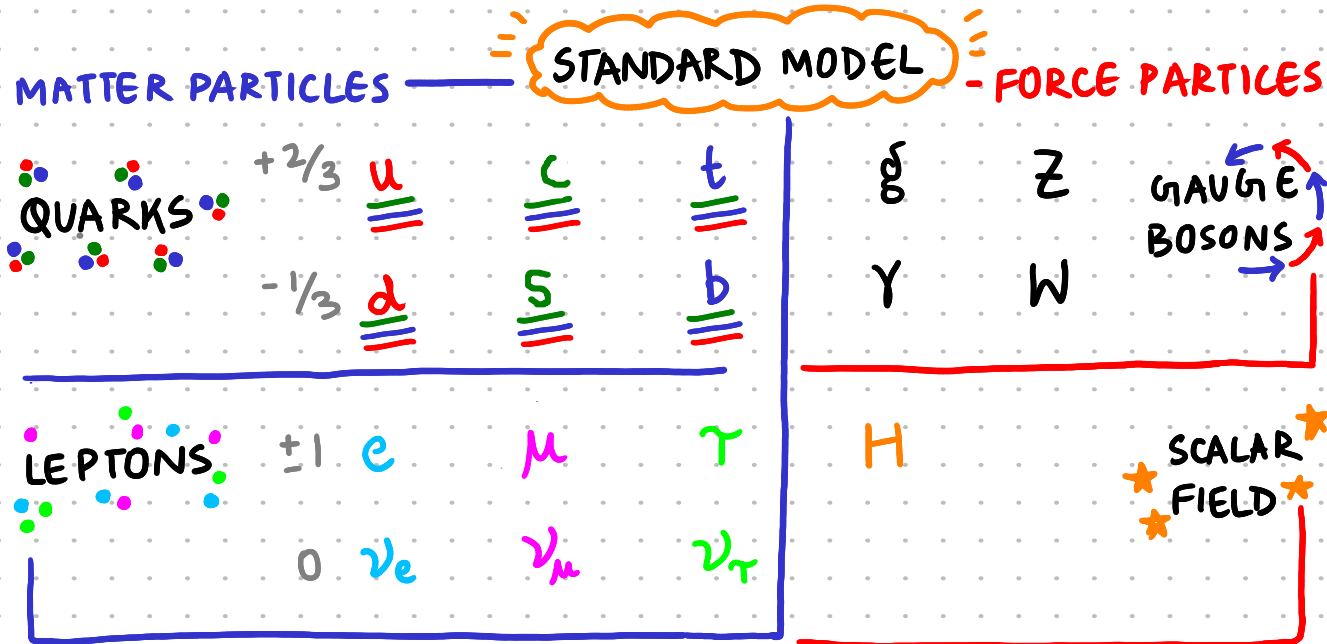




STANDARD MODEL OF PARTICLE PHYSICS

UNIVERSE = made up of MATTER/ENERGY that interacts via FORCES

EVERYTHING
IS A PARTICLE!
(AND A WAVE!)



?
gravity?

dark energy?

?
dark matter?

?
new physics?

! BEYOND STANDARD MODEL !

What happens next?

Is the standard model the best we have?

S.M. IS
CORRECT

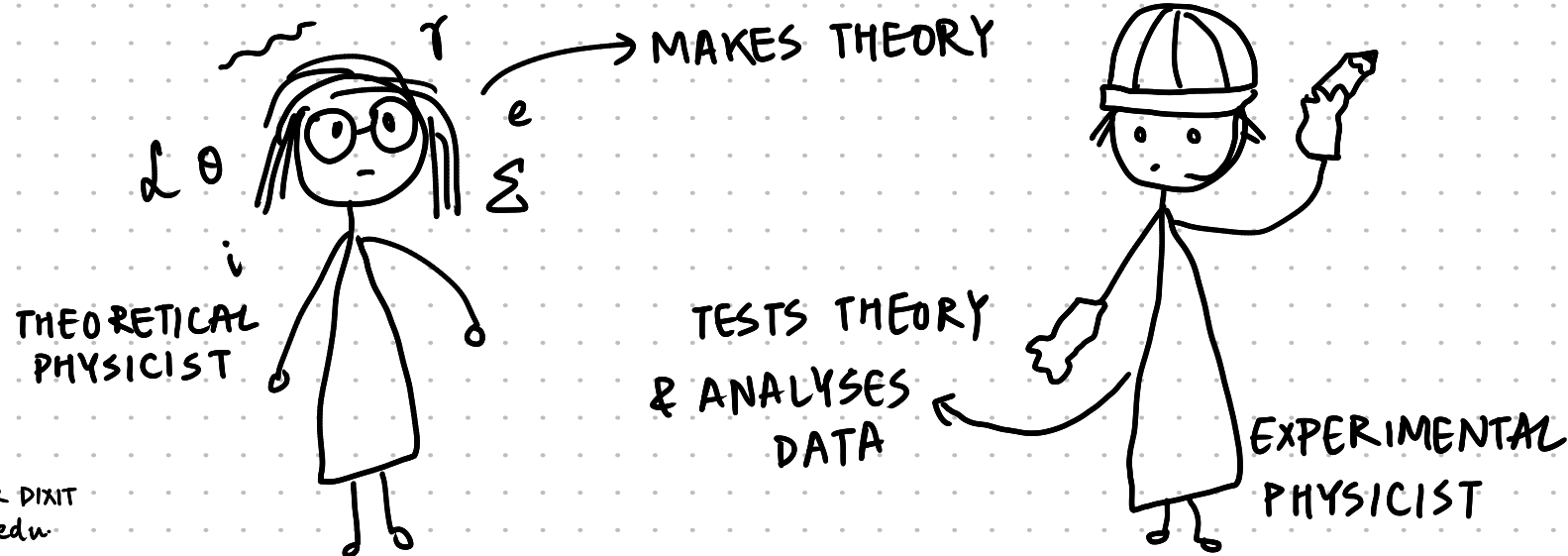
YES

NO

KEEP LOOKING
FOR A BETTER
THEORY.

DO S.M. PARTICLES
HAVE PROPERTIES AS
PREDICTED?

We can test the standard model!



What do we not know yet?

PHYSICS BEYOND THE STANDARD MODEL (BSM)

